

Installing Python and running scripts

Module 1 - Lesson 1

Overview

Python is an interpreted language: the python interpreter reads and runs your source files. This lesson covers installation, running your first program, and the tooling you will use every day.

Key Concepts

- Install Python 3 from python.org or your package manager; check with `python --version`.
- Scripts are plain text files with a `.py` extension; run them with `python hello.py`.
- The interactive REPL is perfect for quick experiments: type `python` and code directly.
- The `print()` function sends output to the terminal.
- Virtual environments isolate project dependencies with `python -m venv .venv`.
- A package manager (`pip`) installs third-party libraries; `requirements.txt` records them.

Example

```
# hello.py
name = input("What is your name? ")
print(f"Hello, {name}!")

# run it
$ python hello.py
What is your name? Ada
Hello, Ada!
```

Summary

Python turns plain-text scripts into programs through its interpreter. A virtual environment and `pip` keep each project's dependencies isolated and reproducible.

Practice Checklist

- Install Python and confirm the version from a terminal.
- Write a script that asks for your name and greets you.
- Create a virtual environment and install a library into it.